

HARSHA DATTATREYA HEGDE

Mangalore, Karnataka, India

+91-7026098039 | harshahegde10s@gmail.com | linkedin.com/in/harsha-hegde-a22233293 | github.com/harshahegde21 | harshahegde.netlify.app

Education

Sahyadri College of Engineering and Management

2023 – Present

Bachelor of Engineering, Computer Science | CGPA: 8.92 / 10.0

Mangalore, Karnataka

- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming

Experience

Backend Developer Intern

January 2026 – Present

Krytil Pvt. Ltd.

Remote

- Optimized MongoDB queries using indexing strategies, improving data retrieval speed by **40%** and reducing average query execution time across critical API endpoints.
- Reduced overall database load by **25%** through targeted query tuning, strategic denormalization, and efficient data access patterns.
- Resolved critical performance bottlenecks in backend services, increasing API throughput by **30%** and improving overall system reliability and uptime.
- Designed and implemented scalable REST API endpoints using Node.js and Express.js following MVC architecture and industry best practices.

Projects

npm Package *Node.js, JavaScript, npm, Semantic Versioning, Open Source*

2025

- Developed and published a JavaScript utility package to the npm ecosystem, accumulating **140+ downloads**, covering the full open-source distribution lifecycle.
- Implemented semantic versioning, dependency management, and comprehensive documentation, enabling seamless integration for JavaScript developers.

Leet Track – LeetCode Problem Tracker | [Live Demo](#) | *React.js, Node.js, Express.js, MongoDB, Vite*

2024

- Architected a full-stack web application for tracking and categorizing LeetCode problems with topic filtering, progress analytics, and RESTful APIs backed by MongoDB.
- Designed a responsive glassmorphism UI with dark/light theme toggle using Tailwind CSS, improving accessibility and user engagement across devices.
- Implemented secure CRUD operations, personalized problem management, and real-time progress tracking to streamline coding interview preparation.

Chats App – Real-Time Messaging Platform *React.js, Node.js, Socket.IO, JWT, MongoDB*

2024

- Built a scalable real-time chat application with one-to-one and group messaging using Socket.IO and WebSockets for low-latency communication.
- Implemented JWT-based authentication and secure REST APIs with MongoDB integration for user management, chat persistence, and reliable communication.
- Deployed the application on Vercel and Render, enabling secure cloud access and real-time communication across devices.
- Live Demo:** chat-s-app.vercel.app

AI Resume Analyzer *React.js, Node.js, LangChain, Ollama, REST API*

2024

- Developed an AI-powered platform using LangChain and Ollama that evaluates resumes against job descriptions, delivering automated relevance scores and skill gap analysis.
- Integrated LangChain prompt chaining to generate structured JSON feedback rendered in a React.js dashboard, significantly reducing manual recruiter screening effort.

Technical Skills

Languages: Java, Python, JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Vite, Tailwind CSS, Responsive Web Design

Backend: Node.js, Express.js, REST API Design, Socket.IO, JWT Authentication, LangChain, Ollama

Databases: MongoDB, MySQL, Database Indexing, Query Optimization, Schema Design

Tools & Platforms: Git, GitHub, Postman, npm, Vercel, Render

Concepts: Data Structures and Algorithms, Object-Oriented Programming, MVC Architecture, WebSockets, Agile

Problem Solving and Open Source

- Solved **350+ DSA problems** on LeetCode and competitive programming platforms, demonstrating strong proficiency in algorithm design, time complexity, and space complexity optimization.
- Contributed to multiple production-grade open-source projects by implementing full-stack logout functionality, improving authentication UX, and integrating Gemini-powered AI features ([Certificate Proof](#)).